



WORK SHEET - 09

COMPLEX NUMBER

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 For any complex number $w = a + bi$, where $a, b \in \mathbb{R}$.

If $w = \cos 40^\circ + i \sin 40^\circ$, then $|w + 2w^2 + 3w^3 + \dots + 9w^9|^{-1}$ equals

- (A) $\frac{1}{9} \sin 40^\circ$ (B) $\frac{2}{9} \sin 20^\circ$ (C) $\frac{1}{9} \cos 40^\circ$ (D) $\frac{9}{2} \operatorname{cosec} 20^\circ$

Q.2 If $x^2 + x + 1 = 0$ then the numerical value of ;

$$\left(x + \frac{1}{x}\right)^2 + \left(x^2 + \frac{1}{x^2}\right)^2 + \left(x^3 + \frac{1}{x^3}\right)^2 + \left(x^4 + \frac{1}{x^4}\right)^2 + \dots + \left(x^{27} + \frac{1}{x^{27}}\right)^2 =$$

- (A) 54 (B) 36 (C) 27 (D) 18

Q.3 The solutions of the equation

$$z^4 + 4z^3i - 6z^2 - 4zi - i = 0$$

are the vertices of a convex polygon in the complex plane. The area of the polygon is

- (A) $2^{3/4}$ (B) $2^{3/2}$ (C) $2^{5/4}$ (D) 2

Q.4 If z be a complex number such that $|z - \alpha^2| + |z - 4\alpha| = 5$, where $\alpha \in \mathbb{R}^+$ always represents an ellipse then the number of integral values of α , is

- (A) 2 (B) 3
(C) 4 (D) 5

Q.5 If z lies on the curve $\arg(z+i) = \frac{\pi}{4}$, then the minimum value of $|z+4-3i| + |z-4+3i|$ is

[Note : $i^2 = -1$]

- (A) 5 (B) 10 (C) 15 (D) 20

Q.6 Let x_1 and x_2 are the roots of the equation $x^2 - x + 1 = 0$ and $\alpha = x_1^{2008} + x_2^{2008}$ and $\beta = x_1^{2009} + x_2^{2009}$ then which one of the following is correct?

- (A) $\alpha = 1$ and $\beta = -1$ (B) $\alpha = -1$ and $\beta = 1$
(C) $\alpha = -1$ and $\beta = -1$ (D) $\alpha = 1$ and $\beta = 1$

Q.7 Let $Z = (\cos 12^\circ + i \sin 12^\circ + \cos 48^\circ + i \sin 48^\circ)^6$ then $\operatorname{Im}(z)$ is equal to

- (A) 0 (B) 1 (C) 3 (D) 5



- Q.8 If $\arg(z+3) = \frac{\pi}{6}$ and $\arg(z-3) = \frac{2\pi}{3}$, then $\tan^2(\arg z) - 2 \cos(\arg z)$, is
 (A) 0 (B) 1 (C) 2 (D) 3
- Q.9 Let z be a complex number. If z satisfies $|z|=1$ and $z=1-\bar{z}$, then principal argument of z is (are)
 (A) $-\frac{\pi}{6}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{3}$ (D) $-\frac{\pi}{3}$
- Q.10 If a complex number $z=x+iy$ ($x, y \in \mathbb{R}$ and $i = \sqrt{-1}$) satisfies the equation $z + \sqrt{2}|z+1| + i = 0$, then $|z|$ is equal to
 (A) $\sqrt{5}$ (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

[MULTIPLE CORRECT CHOICE TYPE]

Q.11 to Q.14 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.11 If $|2z - \omega| = 5$, $|z + 2\omega| = 10$ and $|z + \omega| = 7$, where z and $\omega \in \text{complex numbers}$, then
 (A) $|z| + 2|\omega| = 10$ (B) $|z| + 3|\omega| = 15$
 (C) $2|z| + 3|\omega| + |3 + 4i| = 23$ (D) $2|z|^2 - |\omega|^2 = 23$
- Q.12 If α and β are two distinct complex numbers satisfying $|\alpha|^2 \beta - |\beta|^2 \alpha = \alpha - \beta$ then
 (A) $\arg\left(\frac{\alpha}{\beta}\right) = \pi$ (B) $\alpha\bar{\beta} = \beta\bar{\alpha}$ (C) $\alpha\bar{\beta} = 1$ (D) $|\alpha| = |\beta|$
- Q.13 If $z_1, z_2 \in \mathbb{C}$ such that $z_1 = 5 + 12i$ and $|z_2| = 4$, then
 (A) maximum value of $|z_1 + iz_2| = 17$. (B) minimum value of $|z_1 + (1+i)z_2| = 13 - 4\sqrt{2}$.
 (C) maximum value of $\left| \frac{z_1}{z_2 + \frac{4}{z_2}} \right| = \frac{13}{3}$. (D) minimum value of $\left| \frac{z_1}{z_2 + \frac{4}{z_2}} \right| = \frac{13}{4}$.
- Q.14 If the equation $|z|(z+1)^8 = z^8|z+1|$ where $z \in \mathbb{C}$ and $z(z+1) \neq 0$ has distinct roots $z_1, z_2, z_3, \dots, z_n$ (where $n \in \mathbb{N}$) then which of the following is/are true?
 (A) $z_1, z_2, z_3, \dots, z_n$ are concyclic points. (B) $z_1, z_2, z_3, \dots, z_n$ are collinear points
 (C) $\sum_{r=1}^n \operatorname{Re}(z_r) = \frac{-7}{2}$ (D) $\sum_{r=1}^n \operatorname{Im}(z_r) = 0$

[MATCHING LIST TYPE]

Q.15 & Q.16 are **Matching List type questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Q.15 Let z be a complex number such that $|z - 1| = 2$ and $z' = \frac{3}{3 - \frac{4}{\bar{z} - 1}}$ always lies on a circle whose centre is (x_0, y_0) and radius is r .

List - I

- (P) The value of $(x_0 + y_0)$ is equal to
- (Q) The value of r is equal to
- (R) Maximum value of $|z' + 1|$ is
- (S) Minimum value of $|z'|$ is

List - II

- (1) $\frac{3}{5}$
- (2) $\frac{6}{5}$
- (3) $\frac{9}{5}$
- (4) 4

Codes: P

Q

R

S

- (A) 3 1 2 4
- (C) 4 1 2 3

P

Q

R

S

- (B) 1 3 4 2
- (D) 3 2 4 1

Q.16

List I

- P. Let z, ω, α be complex numbers such that $|z| = |\omega| = 4$ and $\alpha = \frac{z - \bar{\omega}}{16 + z\bar{\omega}}$, then $\text{Re}(\alpha)$ is equal to
- Q. If $x = p + iq$ is a complex number such that $x^2 = 3 + 4i$ and $x^3 = 2 + 11i$ where $i = \sqrt{-1}$, then $(p + q)$ is equal to
- R. Number of complex numbers z satisfying the equation $\bar{z} = iz^2$, where $i = \sqrt{-1}$ is equal to
- S. If $z \in \mathbb{C}$ satisfies $|z + 2 - i| = 5$ then the maximum value of $\frac{|3z + 9 - 7i|}{4}$ is equal to

List II

1. 0
2. 3
3. 4
4. 5

Codes: P

Q

R

S

- (A) 4 3 2 1
- (C) 1 2 3 4

P

Q

R

S

- (B) 1 2 4 3
- (D) 2 1 3 4



[INTEGER TYPE]

Q.17 to Q.20 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

Q.17 Let z_1 and z_2 be two complex numbers such that $|z_1| = 1$ and $|z_2| = 10$. If $\theta = \arg\left(\frac{z_1 - z_2}{z_2}\right)$

then maximum value of $\tan^2\theta$ can be expressed as $\frac{m}{n}$ (where m and n are coprime), find the value of $(100m - n)$.

Q.18 Given that z is a non zero complex number, $iz^2 = 1 + \frac{2}{z} + \frac{3}{z^2} + \frac{4}{z^3} + \frac{5}{z^4} + \dots + \infty$ and $z = n \pm \sqrt{-1}$, then find the value of $(100n)$.

Q.19 Let α^k where $k = 0, 1, 2, \dots, 2013$ are 2014th root of unity. If z_1 and z_2 be any two complex numbers such that $|z_1| = |z_2| = \frac{1}{\sqrt{1007}}$, then find the value of $\sum_{k=0}^{2013} |z_1 + \alpha^k z_2|^2$.

Q.20 Let $z \in \mathbb{C}$ satisfies the equation $\left| \frac{z + \bar{z}}{2 \operatorname{Re}(z)} \right| - |z| = \frac{2}{|\bar{z}|} - \left| \frac{z - \bar{z}}{\operatorname{Im}(z)} \right|$.

If locus of z is curve C_1 or C_2 (C_1 lies in C_2) and chord AB of curve C_2 touches C_1 and from A and B two tangents are drawn to C_1 which meet at C lying on C_2 and if area of $\triangle ABC = \sqrt{k}$, then find the value of $\left\lceil \frac{k}{4} \right\rceil$.

[Note : $[y]$ denotes greatest integer less than or equal to y .]

ANSWER KEY

Q.1	B	Q.2	A	Q.3	C	Q.4	B	Q.5	B
Q.6	B	Q.7	A	Q.8	C	Q.9	CD	Q.10	A
Q.11	AD	Q.12	BC	Q.13	ABC	Q.14	BCD	Q.15	D
Q.16	C	Q.17	[1]	Q.18	[100]	Q.19	[4]	Q.20	[6]

